

INTEGRATED DISEASE MANAGEMENT IN CHILLI (*CAPSICUM ANNUUM*)

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ABSTRACT

Chilli (*Capsicum annum*) is one of the major cash crops in Sri Lanka. However, pests and diseases are found to be the major production constraints of this crop. Researchers have shown that the reduction of agrochemical use can increase the profitability of this crop. In this context, studies were carried out in *yala* 1995, *maha* 1995/96, *yala* 1997 and *maha* 1997/98 seasons to investigate an economically viable integrated method to control major chilli diseases such as reduction of seedling emergence due to damping-off (a nursery disease), chilli leaf curl complex (CLCC), anthracnose and *Cercospora* leaf spot (CLS) and minimize the use of pesticides used in chilli cultivation. The experiments were carried out at Field Crops Research and Development Institute, Maha Illuppallama. Solarization using a wooden frame with double-layered transparent polythene, and seed treatment separately with orthocide 50% (Captan 50 WP) and thiram 80% (Thiram 80 WP) alleviate seedling emergence while burning with paddy husks increased the seedling emergence significantly ($P=0.05$), compared to the untreated control and black polythene treatment. Polythene mulch reduced the disease incidence of CLCC compared to *Gliricidia* and paddy straw mulches. Maize (*Zea mays*) as barrier crop also reduced CLCC, probably due to insect repelling properties. Polythene mulch was effective as insecticides sprayed at 7 day intervals, in reducing CLCC. Varietal screening studies showed that 'Arunalu', Chinese and ANK-31 were resistant to CLCC. Varieties 'Arunalu' and BL-44 were also resistant to anthracnose disease. Of the fungicides tested, Sandoz (cuprous oxide) was found to be superior in controlling anthracnose disease in pods. It appears that these component technologies could be appropriately integrated to alleviate seed emergence, CLCC, anthracnose and CLS in chilli. Integrated pest management (IPM) programme adopted in the farmers' fields revealed that the agrochemical used can be cut down by 50% without affecting the yield.

KEY WORDS: Chilli diseases, Integrated disease management, Solarization, Varietal resistance

INTRODUCTION

Chilli (*Capsicum annum*) is one of the major cash crops in Sri Lanka. Infestation by pest and diseases is the major production constraint of this crop. Chilli leaf curl complex (CLCC) is one of the problems and the major causal factor is *Scirtothrips dorsalis*. Besides, the direct damage of the thrips, and viral diseases transmitted by thrips are found to be important in the *yala* season. Furthermore, chilli diseases such as anthracnose and poor seedling emergence mainly due to damping-off also has caused serious crop losses qualitatively as well as quantitatively during *maha* season. Infection of *Cercospora* leaf spot (CLS) may also result in severe defoliation under humid weather condition.

It has been reported that soil solarization reduces the disease caused by many soil borne pathogens including nematodes, fungi and bacteria (Elad *et al.*, 1980; Grinstein *et al.*, 1979). Polythene used as a mulch may also act as a pest repellent, most probably due to its light reflection properties.

Use of suitable barrier crops to prevent flight activity of the pest towards the crop is another method to minimize pest damage. Maize performed well as a barrier to thrips and thereby controlled CLCC. Efficacy of fungicide use to control the targeted pest or disease is important to obtain economical advantage.

The above cultural practices, application of fungicides and use of resistant varieties to major diseases could be successfully incorporated into an integrated pest management (IPM) programme, which provides a permanent solution.

This study was undertaken to identify individual pest and disease management practices separately and to appropriately integrate them depending on the availability of resources to control diseases in chilli cultivation.

MATERIALS AND METHODS

Nursery treatment

Preparation of polythene frame

Wooden frame was prepared as shown in Fig.1. Transparent polythene was fitted onto both sides of the frame, and outer polythene lining was extended from the edges of the frame to cover the sides of the nursery bed.

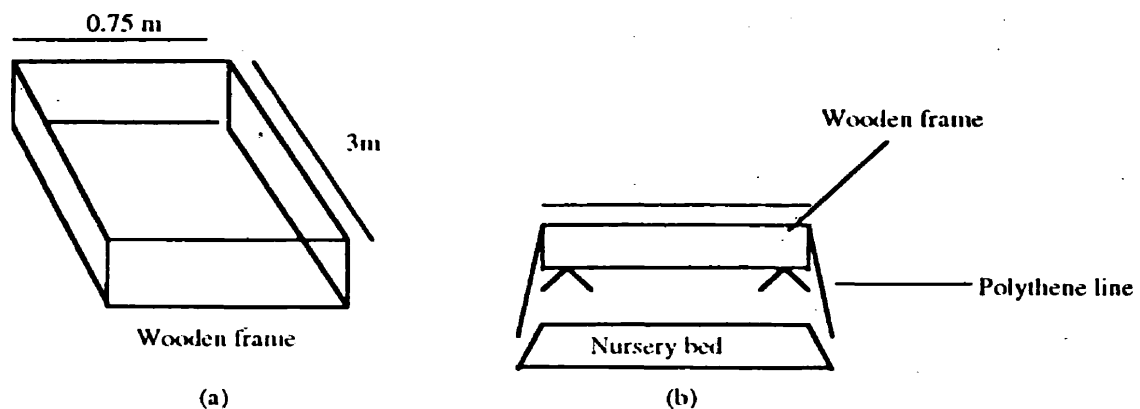


Fig.1. (a) Dimension of wooden frame (b) Nursery set-up of polythene frame

Preparation of nursery bed

The top soil of the nursery bed was loosened to about 4-5cm depth and then a small quantity of water was sprinkled onto the soil. Supporters were placed at the four corners of the frame (Fig.1) in order to raise the frame 5-6 cm above the nursery bed. Nursery bed was covered completely with the extended parts of the outer polythene sheet and the lower edges were sealed with soil. These settings were kept on the nursery beds for one week exposing to bright sunlight in *yala* 1995.

Effect of agrochemicals and other methods on seed emergence of chilli

Nursery beds were prepared using the dimension (3m x 75m x 15m) recommended by the Department of Agriculture (DOA) and the following treatments were arranged in a randomized complete block design (RCBD) with 3 replications. The treatments were seed treatment with Captan 50 WP (6g/kg seeds), seed treatment

with Thiram 80 WP (2g/kg seeds), soil drenching with Morute 80 WP (6g/m²), burning of paddy husks over the seed bed, solarization of nursery beds using a frame fitted with transparent polythene, and black polythene. A nursery bed without any treatments served as the control.

Use of different polythene covers for nursery bed sterilization

Four wooden frames were prepared as follows: The transparent polythene lined on opposite sides of the frame as double layers, black polythene lined on opposite sides of the frames as double layers, transparent polythene lined only one side of the frame as a single layer and black polythene lined on opposite sides of the frames as a single layer. The nursery beds without polythene frame served as the control. Nursery beds were covered with these frames and temperature on the top soil was measured in 30 min intervals from 10.00 am to 4.00 pm for 4 days.

Chilli (cv. MI-2) seedlings were planted at 45cm x 60cm spacing in 3m x 4.5m micro-plots in four replications in the following experiments.

Screening varieties for disease resistance

The varietal screening experiments were carried out at the research field of FCRDI, Maha Illuppallama, under natural infection to identify disease resistant varieties. The plots were arranged in RCBD design with three replicates.

Varietal screening for anthracnose

Fourteen varieties were tested under natural conditions for anthracnose infection during *maha* 1995/96. Varieties MI-1 and MI-2 were used as controls. The disease severity was measured in the pods according to the following scale described in FAO protocols (Anon, 1991).

- 0 - no spots
- 1 - only one or two small circular sunken spots per fruit
- 3 - two or three sunken spots, of about 1cm² in size
- 6 - several bigger spots, more than 1cm² in size, discoloured and rotting
- 9 - more than half of entire surface of the fruit discoloured, black and rotting

The disease severity assessments were taken, 8 weeks after transplanting using naturally infected pods. Disease severity (%) was computed from the following formula:

$$DS(\%) = \frac{(W \times N_s)}{(T \times N_m)} \times 100$$

where, DS = Disease severity; W = Number of wilted plants; N_s = Severity scale; T = Total number of observations; and N_m = Maximum scale number (9).

Varietal screening for CLCC

Twenty eight varieties of chilli were tested for CLCC infection during *yala* 1996. Assessments were done eight weeks after transplanting of chilli seedlings and 0-2 disease scale was used, where 0 - No disease, 1- Mild (< 50 % of the canopy), and 2 - Severe (> 50 % of the canopy). Disease severity (%) was computed by the same formula used for anthracnose.

Fungicidal screening for chilli diseases

Plants were allowed to infect naturally, under field conditions during *maha* 1995/96 at FCRDI, Maha Illuppallama. In the experiments described below, RCBD design was used with four replications. The plot size was 3m x 4.5m. Water sprayed treatment served as control.

Effect of fungicides on anthracnose

To evaluate efficacy of fungicidal formulations against chilli anthracnose, the DOA recommended dosages of the fungicides namely Benomyl 50% WP (Benlate), Orthocide 50% WP (Captan), 73.8% Mancozeb + 6.2 % Carbendazim (Delsene-MX), Copper 50% WP (Sandoz), Propineb 70% WP (Antracol) and Mancozeb 56% + Oxadixyl 10% WP (Sandofan - M) were sprayed at 7-10 day intervals on chilli (cv MI-2).

Effect of fungicide on Cercospora leaf spot

In another experiment, DOA recommended dosages of fungicides, Chlorothalonil 75 WP (Daconil), Thiophenate-methyl (Topsin - M), Orthocide 50% WP (Captan), Orthocide 50% WP (Captan) and Benomyl 50% WP (Benlate) mixture, and Benomyl 50% WP (Benlate) were sprayed to control *Cercospora* leaf spot disease (CLS). The disease assessments were done 72 days after planting.

Effect of amendments and maize barrier crop on chilli diseases

In one treatment 3 rows of maize were sown around the border of micro-plots in *yala* 1996. Chilli plants were established at the time of sowing of maize. Paddy straw and *Gliricidia* leaves were used as separate treatments as amendments on the plots at the rate of 5t/ha. In another treatment transparent polythene was used as mulch. The chilli plots sprayed with only water, and a mixture of Dimethoate (3ml/l) and Sulphur (3g/l) served as controls.

Effect of IPM programme on the yield

Based on the results of the above experiments the adaptation of IPM was carried out in farmers' fields at Maha Illuppallama in *yala* 97 and *maha* 1997/98 (Table 1).

Table 1. IPM programme adopted during yala 1997 and maha 1997/98

Stage	Operations	
	yala 1997	maha 1997/98
Variety	Arunalu	Arunalu
Nursery level	Soil solarization using transparent double polythene frame	Burning with paddy husk followed by soil drenching of fungicides
Field level	Polythene mulch to control CLCC followed by insecticidal sprays	Fungicidal sprays commenced at the initial stage of the anthracnose symptoms

Plot size was 10m x 5m. Plots where IPM was not adopted were sprayed 6 and 10 times with pesticides in yala and maha seasons, respectively, whereas IPM adopted trials were sprayed only 3 and 5 times. The total dry chilli yield and marketable yield were recorded.

All experiments described above were repeated at least twice and the other practices such as fertilizer application, nursery preparation and irrigation were carried out according to the DOA recommendation.

RESULTS

The percentage seedling emergence was significantly higher in all treatments compared to the control (Table 2). Soil solarization with double-layered black polythene frame showed significantly lower seedling emergence than other treatments.

However, there were significant differences among seed treatment with captan or thiram, burning of paddy husks or soil solarization with double layered transparent polythene cover.

Table 2. Emergence of chilli seedlings after different treatments

Treatment	Emergence (%)	
	18 DAS	25 DAS
Black polythene frame	60.8 (51.3) c	61.9 (52.0) c
Transparent polythene frame	83.4 (66.0) a	84.8 (67.1) a
Burning of paddy husk	79.9 (63.4) ab	82.7 (65.6) a
Seed treatment with captan	83.2 (66.0) a	86.6 (68.6) a
Seed treatment with thiram	82.3 (65.2) a	80.9 (64.2) a
Soil drenching with morute	73.7 (59.2) b	71.0 (57.5) b
Control (without treatment)	37.2 (37.6) d	36.4 (37.1) d

* Each figure is the mean of three replications; Means followed by a common letter are not significantly different at 5% level. Arcsine transformed data are within parentheses; DAS = Days after sowing

The maximum temperature (54°C) was achieved between 2.00 pm and 3.00 pm when double-layered polythene frame was used for soil solarization (Fig.2). The black polythene frames did not show any increase of soil temperature compared to control.

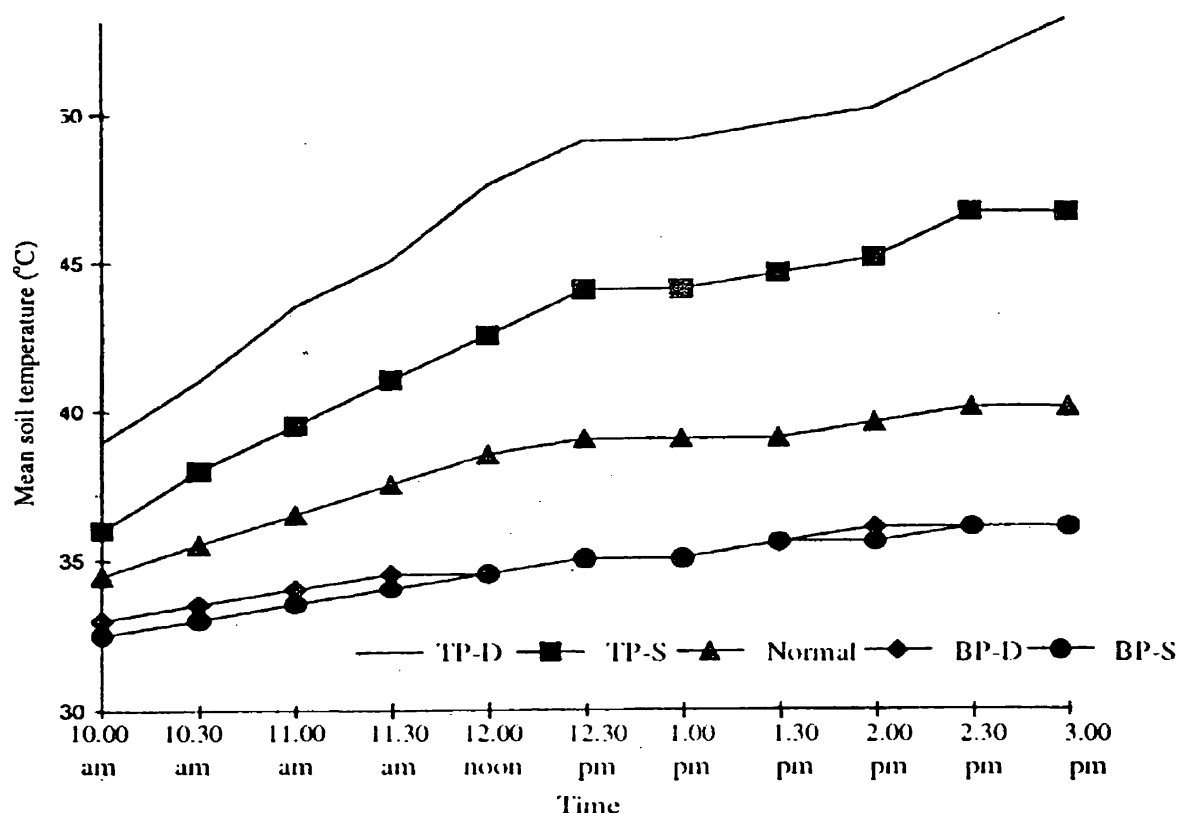


Fig. 2. Comparison of soil temperature build-up under different polythene frames used to cover the nursery beds (Normal = Control; TP-D = Transparent polythene double layer; TP-S = Transparent polythene single layer; BP-D = Black polythene double layer; BP-S = Black polythene single layer)

The experiment on natural infection of anthracnose showed that varieties 'Arunalu' and BL44 possessed low disease severities of anthracnose in pods (Table 3).

Table 3. Disease severity in chilli varieties caused by anthracnose disease in pods, incited by *Colletotrichum capsici* and *Colletotrichum gloeosporioides* under natural infection at FCRDI, Maha Illuppallama during maha 1995/96

Variety	Disease severity (%)*
PVT 10	22.6 (29.0) a
MI-2	16.7 (12.3) b
ANK-11	12.9 (20.4) c
BL-20	11.4 (20.3) c
PC-1	7.0 (15.9) d
BL 52	6.8 (15.7) d
ANK-8	6.4 (14.9) de
ANK-2	6.2 (13.7) def
PVT 22	4.0 (12.4) efg
MI-1	4.0 (12.4) efg
ANK-6	3.9 (12.3) efg
PVT 20	2.6 (9.6) gh
Arunalu	1.7 (7.9) h
BL-44	1.4 (7.1) h

* 0-9 scale was used to assess disease severity (Please see text). Each value is the mean of three replications. Means followed by a common letter are not significantly different at 5% level. Arcsine transformed data are within parentheses.

In contrast PVT-10 was highly susceptible to anthracnose compared to other varieties.

It was observed that varieties 'Arunalu', Chinese and ANK 31 were resistant to CLCC, whilst variety PVT-1 was highly susceptible to the disease (Table 4).

In general, all fungicides gave acceptable control of anthracnose (Table 5). Antracol is superior when compared with other chemicals in controlling anthracnose in plants. However, it seems that it is ineffective in controlling the disease in pods. Of the fungicides, copper sandoz gave highest disease control in pods. The percentage lesion area in pods of all treatments was found to be not significant. The results seem to show that once the pods get infected the lesion caused by the fungus can attain its maximum size and none of the fungicide was effective against lesion development.

Table 4. Varietal screening for Chilli Leaf Curl Complex (CLCC) during yala 1996 under natural infection at FCRDI, Maha Illuppallama

Variety	Disease severity (%)*
PC-1	76.0 (60.4) a
KA65U	66.8 (55.3) b
KA2T7	62.0 (50.8) c
PS-3	54.2 (47.4) cd
PC-11	53.7 (47.1) cd
PVT-1	53.1 (46.8) d
PS-1	48.0 (42.3) e
PVT-2	44.4 (41.8) e
BL-20	42.3 (40.6) ef
ANK72	40.6 (39.6) efg
PVT-3	40.0 (39.2) efg
ANK121	39.4 (38.9) efg
BL-52 (St-2)	39.1 (38.7) efg
ANK121	38.7 (38.5) efg
PVT-6	36.8 (37.4) fg
PVT-5	35.7 (36.7) fg
MI-1	35.4 (36.5) g
ANK101	29.0 (32.8) h
MI-2	28.5 (32.2) h
PS-2	28.1 (32.1) h
ANK22	27.5 (31.5) hi
BL-44	26.7 (31.2) hi
BL-52 (St-1)	20.8 (28.0) ij
PVT-4	20.8 (26.9) jk
Arunalu	17.6 (24.3) k
Chinese	17.5 (23.7) k
ANK31	12.5 (19.9) l

* 0-2 scale was used to calculate disease severity (Please see text). Each value is the mean of three replications. Means followed by a common letter are not significantly different at 5% level. Arcsine transformed data are within parentheses.

Table 5. Efficacy of fungicidal formulations in controlling of chilli anthracnose, 77 days after planting during *maha* 1995/96 under natural infection at FCRDI, Maha Illuppallama

<i>Fungicide</i>	<i>Dosage (a.i./l)</i>	<i>Diseased plants (%)</i>	<i>Infected pods (%)</i>	<i>Lesion area in pods (%)</i>
Benlate (Benomyl)	0.6 g	21.3	34.6	40.1
Captan (Orthocide)	2.0 g	24.4	27.3	36.5
Delsene-X (Carbendazim + Mancozeb)	2.0 g	28.2	32.1	34.9
Copper sandoz (Cuprous oxide)	4.0 g	25.7	19.1	38.5
Anracol (Propineb)	2.0 g	15.5	39.7	34.6
Sandofan-M (Mancozeb + Oxadixyl)	3.0 g	29.4	28.5	39.5
Control (sprayed with water)		33.1	34.9	35.8
LSD (P=0.05)		1.1	9.6	NS

* Each figure is the mean of four replicates; NS = Not significant

It was observed that the Daconil and the mixture of Captan and Benlate had effectively controlled CLS disease (Table 6).

Table 6. Fungicidal screening against *Cercospora* leaf spot during *maha* 1995/96 under natural infection at FCRDI, Maha Illuppallama

<i>Fungicide</i>	<i>Dosage (g /100 l)</i>	<i>Disease severity (%)*</i>
Daconil	1.75	0.77
Topsin-M	0.75	1.08
Captan	2.00	1.45
Captan + Benlate	1.34 + 0.67	0.98
Benlate	0.60	2.06
Control	-	2.33
LSD (P=0.05)		0.52
CV (%)		23.90

* Each figure is the mean of four replicates

The chilli plots mulched with transparent polythene significantly reduced the disease incidence of CLCC which was almost similar to the insecticidal sprayed control (Table 7). Soil amendment with paddy straw, *Gliricidia* or use of maize as barrier crop did not show significant reduction of CLCC.

The marketable yield showed a remarkable increase in IPM adopted chilli plots during *maha*, compared to non-adopted plots in the same season (Table 8). This is due to improvement of pod quality. However, this pattern was not found during *yala*.

Table 7. Effect of various cultural practices to control Chilli Leaf Curl Complex (CLCC) during yala 1996

Cultural practice	Disease incidence*	
	68 DAP	84 DAP
Maize as barrier crop ^a	57.9	62.2
Paddy straw mulch ^b	63.5	75.8
<i>Gliricidia</i> mulch ^b	62.9	72.1
Polythene mulch	25.9	46.1
Sprayed ^c	29.9	34.5
Unsprayed (Control)	49.0	57.8
LSD (P=0.05)	5.4	7.7
CV(%)	23.2	18.1

* Each value is the mean of four replicates; DAP = Days after planting; a = Three rows of maize cultivated around the border of the plot; b = 10 t/ha; c = Sprayed with Dimethoate (30 ml) and Sulphur (30 g)

Table 8. Effect of adaptation of IPM on the total and marketable yield of chilli during yala 1997 and maha 1997/98

Treatment	yala 1997			maha 1997/98		
	No. of pesticide applications "	Total yield (t/ha)	Marketable yield (t/ha)	No. of pesticide applications "	Marketable yield (t/ha)	Total yield (t/ha)
With IPM	3	3.2	2.8	5	3.5	3.2
Without IPM	6	2.8	2.6	10	3.0	1.8

a = insecticide (Dimethoate 3ml/l) and fungicide (Sandoz 4g/l)

DISCUSSION

The results showed that the nursery sterilisation can be done by solarization using transparent polythene frame. This frame can be used over several seasons if it is stored carefully to avoid damage to polythene. Limitation of this technique is the requirement of abundant sunlight during the period of nursery establishment. Researchers have shown that soil solarization for 4 to 6 week periods at temperatures between 40 – 50°C is sufficient to kill many pathogenic organisms such as *Sclerotium rolfsii*, *Pythium* sp., *Rhizoctonia solani* and *Thielavopsis basicola* (Katan, 1987 ; Pullman *et al.*, 1981). The results in Fig.2 showed that soil solarization using double-layered transparent polythene frame for a period of one week can achieve mean temperatures between 40 – 53°C from 10.30 am to 3.00 pm. Therefore, this may be sufficient to control most of the soil-borne plant pathogens. The transparent double-layered polythene frame also increased seedling emergence probably due to suppression of fungal pathogens (Table 1).

The covering of bed with reflective mulches is a common cultural practice adopted by many vegetable growers in Taiwan to control aphids and thrips transmitted viral diseases (Greenough, 1985). Since, CLCC is a serious problem during yala

season polythene amendment could be used to control CLCC more effectively. The major component of CLCC was thrip (Wijeratne Banda, 1997). Their direct as well as indirect damage to the chilli plants through virus transmission possibly cause marked yield losses. Therefore, polythene amendment coupled with resistant varieties such as ANK 31, Chinese and Arunalu' can effectively reduce the insecticidal sprays against thrips and also the thrip-transmitted viral diseases.

Of the varieties tested, 'Arunalu', Chinese and ANK 31 were resistant to CLCC. 'Arunalu' also showed acceptable resistance to anthracnose disease. Thus, it is suitable to cultivate 'Arunalu' during *maha* season where the disease is found to be prominent. Fungicide Sandoz showed significant control of anthracnose in pods. The effectiveness of these chemicals could be enhanced when the spraying is commenced from the flowering stage.

Therefore, integration of the resistant variety 'Arunalu' and timely spraying with Sandoz will greatly reduce the anthracnose which is the major problem in *maha*. Adaptation of this IPM programme in the farmers' fields revealed that 50% reduction in the agrochemical application improved the marketable yield during *maha*. Therefore, in addition to the reduction of pesticide cost, the quality of chilli pods can also be improved by following the new IPM programme.

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